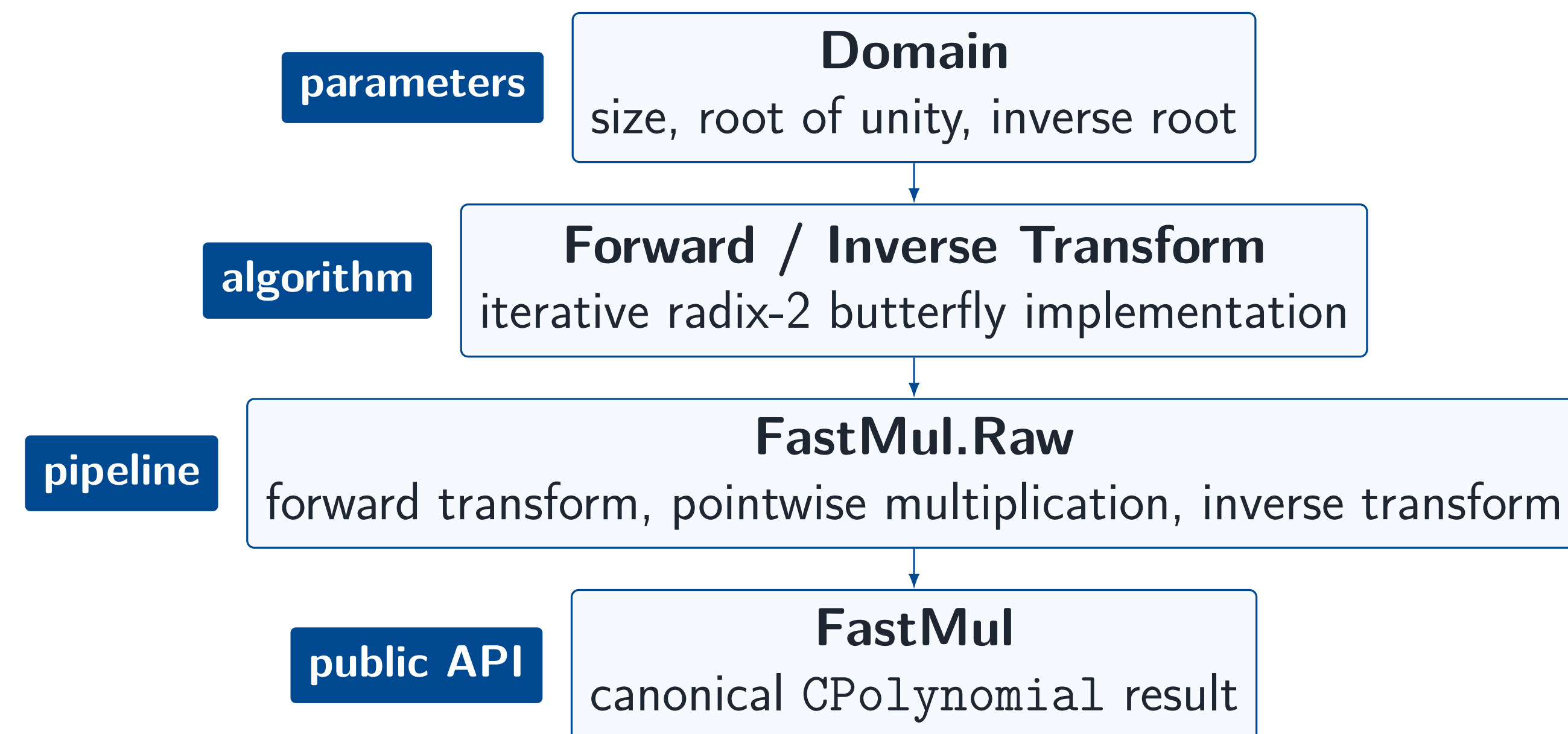


Motivation

Polynomial multiplication is a basic operation behind computable algebra and proof-system code. This project adds a faster path to CompPoly and proves that the executable result is the same polynomial as ordinary multiplication.

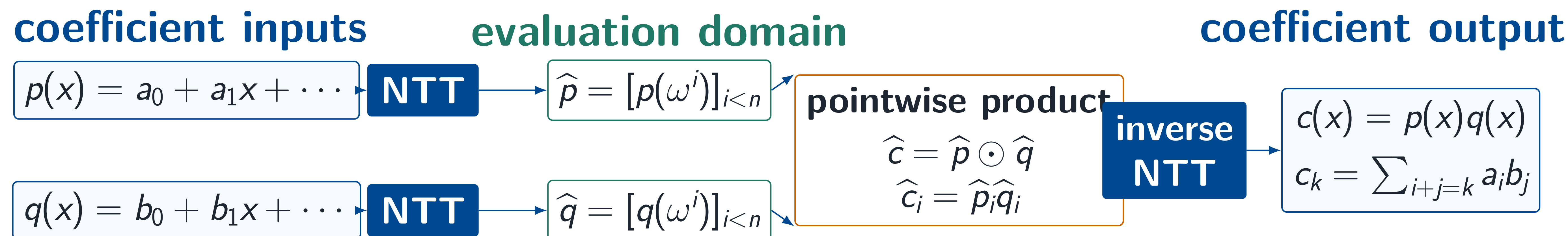
Method	Main operation	Growth
Naive multiplication	coefficient convolution	$O(n^2)$
NTT multiplication	transform, pointwise product, inverse	$O(n \log n)$

Implementation Surface



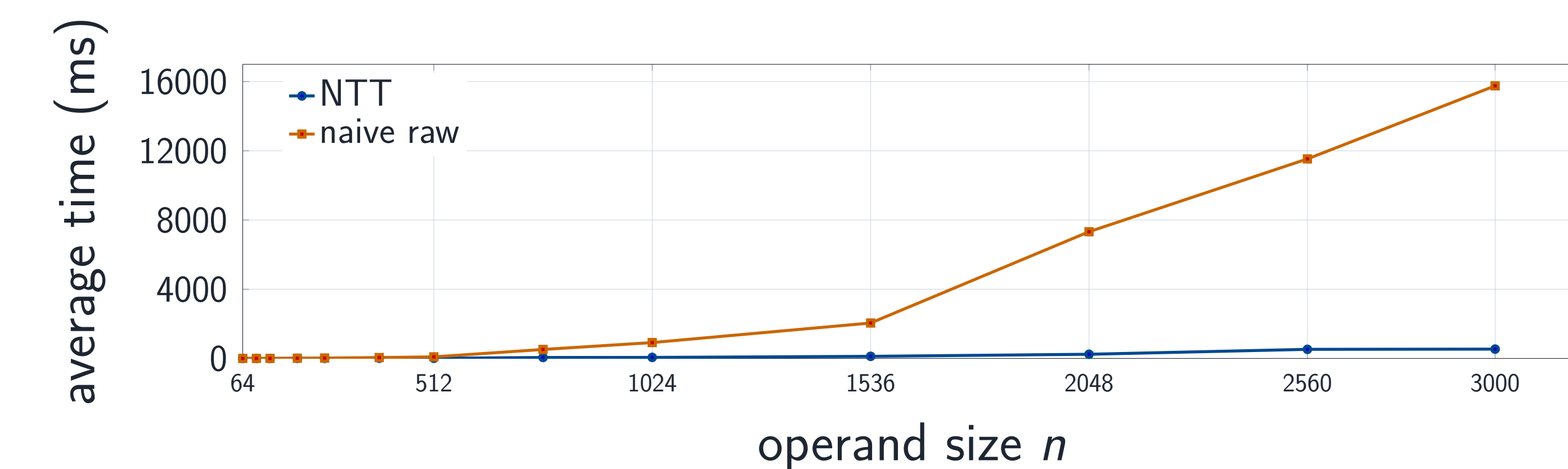
The public operation is small; most of the work is in connecting executable array code to mathematical transform specifications and then to polynomial multiplication.

NTT Multiplication



The transform evaluates at powers of a root of unity. Multiplication becomes pointwise in the evaluation domain, then the inverse transform returns the coefficient representation.

Benchmark Behavior



Manual KoalaBear benchmark sweep from
 tests/CompPolyTests/Univariate/NTT/Benchmark.lean. Times are illustrative and machine-dependent.

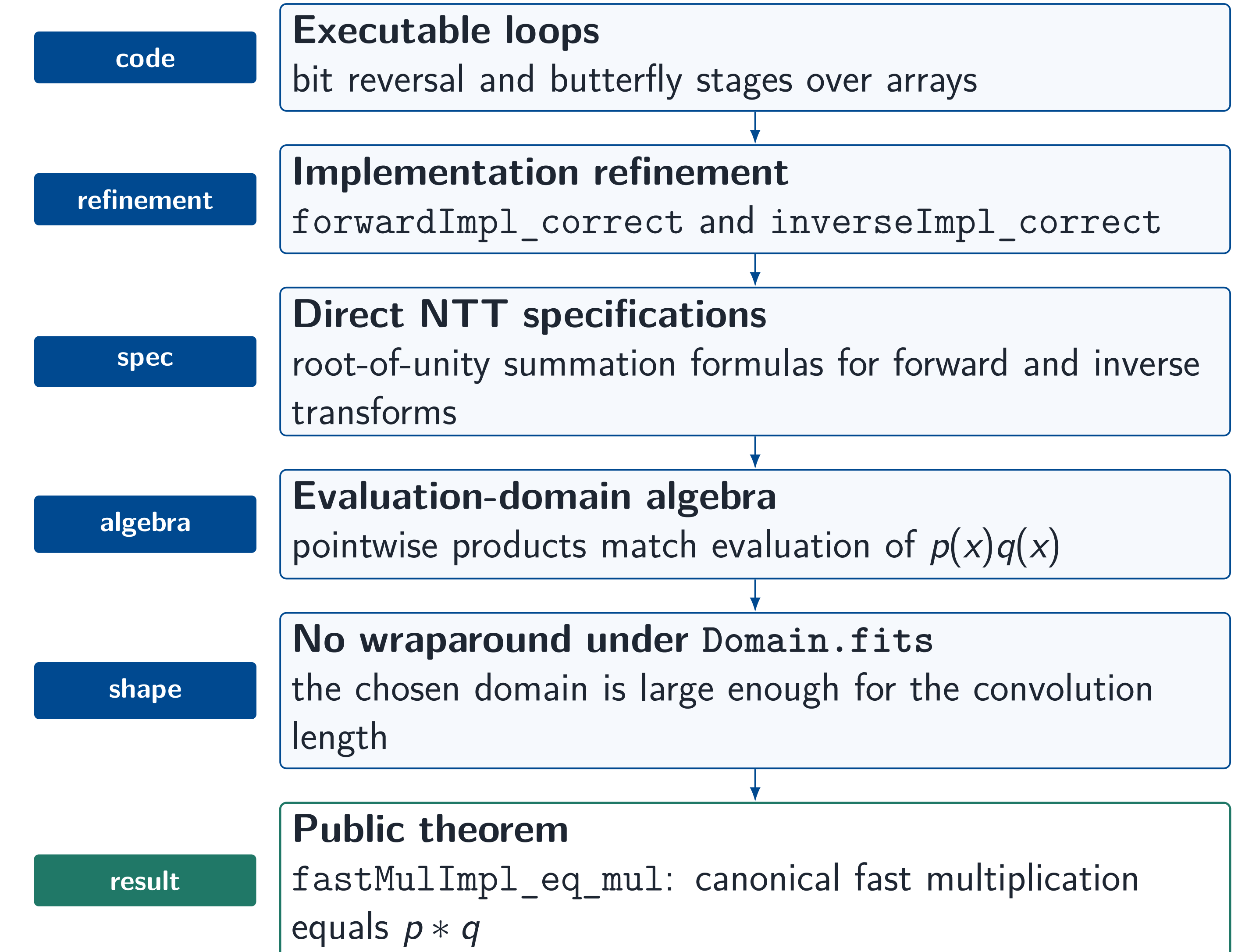
Main Theorem

```

theorem fastMulImpl_eq_mul
  (D : Domain R)
  (p q : CPolynomial R)
  (hfit : Domain.fits D p.val q.val) :
  fastMulImpl D p q = p * q
  
```

This is the public correctness statement for the multiplication implementation. The precondition `Domain.fits D p.val q.val` says that the selected NTT domain is large enough for the ordinary convolution, so the transform pipeline has no cyclic wraparound.

Proof Architecture



References

1. Trieu, A., "Formally Verified Number-Theoretic Transform." *IACR Communications in Cryptology*, vol. 2, no. 4, Jan. 2026. DOI: 10.62056/ahbn-4tw9.
2. Capretta, V., "Certified Fast Fourier Transform." *Theorem Proving in Higher Order Logics*, TPHOLs 2001. Springer, 2001.
3. Ballarin, C., "Fast Fourier Transform." *Archive of Formal Proofs*, Oct. 2005. isa-afp.org/entries/FFT.html.
4. Truth Research ZK, "AMO-Lean", 2026. github.com/lambdaaclass/amo-lean.

Code and Repository



Merged PR #174



CompPoly Repository